



KG-2 BOOK A



Composition of 10 Advanced & Mental Calculation

Student Information

Name:

.....

Level:

.....

Date:

.....

7

8

9



Composition of 10

Each problem rewrites as a two-step trick



$$7+5 : 7-5+10 = 12$$

$$\begin{array}{r} 7 \\ +5 \\ \hline \end{array}$$



$$9+4 : 9-6+10 = 13$$

$$\begin{array}{r} 9 \\ +4 \\ \hline \end{array}$$



$$8+3 : 8-7+10 = 11$$

$$\begin{array}{r} 8 \\ +3 \\ \hline \end{array}$$



Three-Step Problems

 Speedy mascots fly high — solve each problem




$$7 - 3 + 6 =$$


$$9 + 7 - 4 =$$


$$9 + 6 - 1 =$$


$$6 - 3 + 8 =$$


$$8 - 4 + 9 =$$



Abacus Calculation: Composition of 10 (+4)



+4 = -6 + 10 Fingering Ex.: 6+4, 7+4, 8+4, 9+4

S.No.	1	2	3	4	5	6	7	8
1	7	6	8	9	6	7	8	9
2	+4	+4	+4	+4	+4	+4	+4	+4
3	-1	-3	-2	-1	-2	-3	-1	-2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	6	7	8	9	6	7	8	9
2	+4	+4	+4	+4	+4	+4	+4	+4
3	-1	-2	-3	-1	-2	-1	-2	-3
Ans.								

S.No.	1	2	3	4
1	7	8	9	6
2	+4	+4	+4	+4
3	-1	-2	-3	-1
Ans.				

$4 + 6 =$

$9 + 4 =$

$8 + 4 =$

$7 + 4$
 $= 7 - 6 + 10$
 $= 11$

11

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	3	7	9	2	6	8	4	2
2	+5	+4	-3	+6	+8	-2	+5	+6
3	-2	-1	+5	+4	-3	+4	-1	-3
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	5	7	4	5	9	7	6	5
2	-1	-3	+3	+2	-2	-4	-3	+3
3	+4	+8	-5	-4	+3	+2	+4	-2
Ans.								

S.No.	1	2	3	4	☆
1	7	3	4	7	
2	+3	+6	+7	+3	
3	-4	+2	-1	+4	
Ans.					



+


5 + 4 =



+

6 + 4 =



+

7 + 4 =



-

9 - 4 =

Abacus Calculation: Composition of 10 (+3)



+3 = -7 + 10 Fingering Ex.: 7+3, 8+3, 9+3

S.No.	1	2	3	4	5	6	7	8
1	8	7	9	7	8	9	7	8
2	+3	+3	+3	+3	+3	+3	+3	+3
3	-1	-2	-3	-1	-2	-1	-3	-2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	9	7	8	9	7	8	9	7
2	+3	+3	+3	+3	+3	+3	+3	+3
3	-2	-1	-3	-2	-3	-1	-2	-1
Ans.								

S.No.	1	2	3	4
1	8	9	7	8
2	+3	+3	+3	+3
3	-2	-1	-3	-2
Ans.				

+

 $7 + 3 =$

+

 $8 + 3 =$

+

 $9 + 3 =$

$8 + 3$
 $= 8 - 7 + 10$
 $= 11$

11

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8	9	10
1	2	6	7	3	6	8	5	6	6	2
2	+7	-3	-1	+5	-2	+5	+3	+2	+3	+7
3	-5	+7	+3	+6	+4	-3	-6	-4	-5	-3
Ans.										

S.No.	1	2	3	4	5	6	7	8	9	10
1	3	5	5	5	5	1	2	3	4	5
2	-1	-4	-2	+3	+8	+3	+4	+5	+6	+7
3	+5	+7	+4	-4	-3	+8	+3	-2	-5	-6
Ans.										

S.No.	1	2	3	4	5	6	7	8	9	10
1	1	6	7	5	7	3	2	3	4	6
2	+6	+8	-2	+6	+8	+6	+7	-2	+5	+4
3	+2	-7	+8	-3	-3	+2	+5	+8	-3	-2
Ans.										



Abacus Calculation: Revision



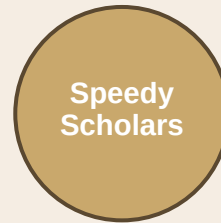
S.No.	1	2	3	4	5	6	7	8	9	10
1	3	4	7	8	6	5	1	6	2	3
2	+6	-2	+5	-3	+8	-2	+4	+7	+7	+8
3	-2	+5	-3	+5	-4	+3	+2	-5	+8	-2
Ans.										

S.No.	1	2	3	4	5	6	7	8	9	10
1	2	6	4	5	7	8	2	6	8	3
2	-1	-3	+8	+5	-3	+4	+3	-2	-4	+4
3	+5	+7	-4	-3	+5	-1	-1	+5	+7	+8
Ans.										

S.No.	1	2	3	4	5	6	7	8	9	10
1	1	2	6	4	5	3	2	4	3	5
2	+4	+6	+4	-1	+6	+7	+8	+5	-2	+7
3	+2	+8	-3	+3	-2	-4	+5	-2	+7	-3
Ans.										

Match the Birds to Form 10

Draw a line between each pair of mascots that sums to 10



Abacus Calculation: Composition of 10 (+2)



+2 = -8 + 10 Fingering Ex.: 8+2, 9+2

S.No.	1	2	3	4	5	6	7	8
1	8	9	8	9	8	9	8	9
2	+2	+2	+2	+2	+2	+2	+2	+2
3	-1	-2	-3	-1	-2	-1	-3	-2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	8	9	8	9	8	9	8	9
2	+2	+2	+2	+2	+2	+2	+2	+2
3	-3	-1	-2	-3	-1	-2	-3	-1
Ans.								

S.No.	1	2	3	4
1	9	8	9	8
2	+2	+2	+2	+2
3	-3	-1	-2	-3
Ans.				

+

 $8 + 2 =$

+

 $9 + 2 =$

+

 $7 + 3 =$

$8 + 2$
 $= 8 - 8 + 10$
 $= 10$

10

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	5	7	8	9	6	7	8	9
2	+4	+3	+2	+1	+4	+3	+2	+4
3	-3	-2	-1	-3	-2	-1	-3	-2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	3	4	5	6	7	8	9	5
2	+7	+6	+5	+4	+3	+2	+1	+4
3	-2	-1	-3	-2	-1	-3	-2	-3
Ans.								

S.No.	1	2	3	4
1	6	7	8	9
2	+3	+2	+4	+3
3	-1	-3	-2	-1
Ans.				




$9 + 1 =$




$8 + 2 =$




$7 + 3 =$




$6 + 4 =$

Abacus Calculation: Composition of 10 (+1)



+1 = -9 + 10 Fingering Ex.: 9+1

S.No.	1	2	3	4	5	6	7	8
1	9	9	9	9	9	9	9	9
2	+1	+1	+1	+1	+1	+1	+1	+1
3	-2	-3	-1	-2	-3	-1	-2	-3
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	9	9	9	9	9	9	9	9
2	+1	+1	+1	+1	+1	+1	+1	+1
3	-1	-2	-3	-1	-2	-3	-1	-2
Ans.								

S.No.	1	2	3	4
1	9	9	9	9
2	+1	+1	+1	+1
3	-3	-1	-2	-3
Ans.				




$9 + 1 =$




$8 + 1 =$




$7 + 1 =$

$9 + 1$
 $= 9 - 9 + 10$
 $= 10$

10

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	9	8	7	6	5	9	8	7
2	+1	+2	+3	+4	+5	+2	+3	+4
3	-3	-1	-2	-1	-3	-1	-2	-3
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	6	5	4	3	2	9	8	7
2	+5	+6	+7	+8	+9	+1	+2	+3
3	-1	-2	-3	-2	-1	-2	-3	-1
Ans.								

S.No.	1	2	3	4
1	6	5	4	3
2	+4	+5	+6	+7
3	-2	-3	-1	-2
Ans.				

















6 + 5 =

7 + 5 =

8 + 5 =

9 + 5 =

Composition of 10

Each visual problem rewrites to a 10-trick



$$7 + 6 : 7 - 4 + 10 = (\quad)$$



$$\begin{array}{r} 7 \\ +6 \\ \hline \end{array}$$

$$7 + 7 : 7 - 3 + 10 = (\quad)$$



$$\begin{array}{r} 7 \\ +7 \\ \hline \end{array}$$



Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	5	6	7	8	9	5	6	7
2	+6	+5	+4	+3	+2	+5	+6	+7
3	-2	-1	-3	-2	-1	-3	-2	-3
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	8	9	5	6	7	8	9	5
2	+5	+6	+7	+8	+9	+4	+3	+5
3	-2	-3	-1	-2	-3	-1	-2	-3
Ans.								

S.No.	1	2	3	4
1	6	7	8	9
2	+7	+8	+9	+5
3	-1	-2	-3	-1
Ans.				





5 + 6 =





6 + 6 =





7 + 6 =





8 + 6 =

Abacus Calculation

Solve every piece of the puzzle and write the answer in the circle



$3 + 8 + 1$		$5 - 2 + 7$		$6 - 4 + 8$		$7 + 2 - 3$	
$4 + 9 - 5$		$3 + 6 - 2$		$8 - 3 + 5$		$9 + 1 - 4$	
$5 + 4 - 2$		$7 - 5 + 9$		$2 + 7 + 3$		$6 + 4 - 8$	



Abacus Calculation: Composition of 10 (+6)



+6 = -4 + 10 Fingering Ex.: 5+6, 6+6, 7+6

S.No.	1	2	3	4	5	6	7	8
1	5	6	7	5	6	7	5	6
2	+6	+6	+6	+6	+6	+6	+6	+6
3	-2	-3	-1	-3	-2	-1	-3	-1
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	7	5	6	7	5	6	7	5
2	+6	+6	+6	+6	+6	+6	+6	+6
3	-2	-1	-3	-2	-3	-1	-2	-3
Ans.								

S.No.	1	2	3	4
1	6	7	5	6
2	+6	+6	+6	+6
3	-1	-2	-3	-2
Ans.				


 +
 

 $5 + 6 =$


 +
 

 $6 + 6 =$


 +
 

 $7 + 6 =$

$5 + 6$
 $= 5 - 4 + 10$
 $= 11$
11

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	5	6	7	8	9	5	6	7
2	+7	+6	+5	+4	+3	+8	+7	+6
3	-2	-3	-1	-2	-1	-2	-3	-1
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	5	6	7	8	9	5	6	7
2	+5	+4	+3	+2	+1	+6	+5	+4
3	-3	-2	-1	-3	-2	-1	-2	-3
Ans.								

S.No.	1	2	3	4
1	8	9	5	6
2	+3	+2	+7	+6
3	-1	-2	-3	-1
Ans.				

+

8 + 5 =

+

9 + 5 =

+

5 + 5 =

+

7 + 5 =

The Little Cinema

Solve each problem and lead each mascot to its numbered seat



$1+3+8$

$2+6-4$

$6+2+8$

$7-5+9$

$3+5+9$

$2+5+8$

$6-2+1$

$8-6+4$

$3+1+3$



Abacus Calculation: Composition of 10 (+7)



+7 = -3 + 10 Fingering Ex.: 5+7, 6+7, 7+7

S.No.	1	2	3	4	5	6	7	8
1	5	6	7	5	6	7	5	6
2	+7	+7	+7	+7	+7	+7	+7	+7
3	-2	-3	-1	-3	-2	-1	-3	-2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	7	5	6	7	5	6	7	5
2	+7	+7	+7	+7	+7	+7	+7	+7
3	-1	-2	-3	-1	-3	-2	-1	-2
Ans.								

S.No.	1	2	3	4
1	6	7	5	6
2	+7	+7	+7	+7
3	-3	-1	-2	-3
Ans.				

 +
 

5 + 7 =

 +
 

6 + 7 =

 +
 

7 + 7 =

5 + 7
 = 5-3+10
 = 12

12

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	5	6	7	8	9	5	6	7
2	+8	+7	+6	+5	+4	+7	+8	+5
3	-1	-2	-3	-1	-2	-3	-1	-2
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	8	9	5	6	7	8	9	5
2	+6	+7	+5	+6	+7	+3	+2	+9
3	-3	-1	-2	-3	-1	-2	-3	-1
Ans.								

S.No.	1	2	3	4
1	6	7	8	9
2	+8	+5	+4	+1
3	-2	-3	-1	-2
Ans.				

+

 $8 + 4 =$

+

 $7 + 5 =$

+

 $6 + 6 =$

+

 $9 + 2 =$

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	5	6	7	8	9	5	6	7
2	+6	+7	+5	+4	+3	+5	+6	+7
3	-2	-3	-1	-2	-1	-2	-3	-1
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	8	9	5	6	7	8	9	5
2	+5	+6	+7	+8	+9	+4	+3	+5
3	-3	-2	-1	-3	-2	-1	-3	-2
Ans.								

S.No.	1	2	3	4
1	6	7	8	9
2	+7	+6	+5	+4
3	-1	-2	-3	-1
Ans.				




$5 + 8 =$




$6 + 8 =$




$4 + 8 =$




$5 + 9 =$

Composition of 10 — Bigger Sums

Two new tricks: +8 and +9 with starts of 5+



$$5 + 8 = 5 - 2 + 10$$



$$\begin{array}{r} 5 \\ +8 \\ \hline \end{array}$$

$$5 + 9 = 5 - 1 + 10$$



$$\begin{array}{r} 5 \\ +9 \\ \hline \end{array}$$



Match the Mascot to its House

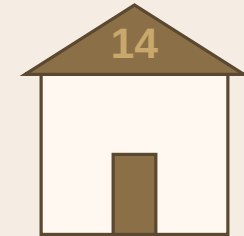
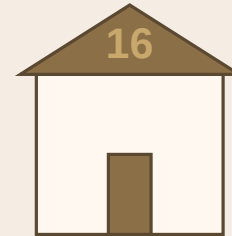
Solve each problem and draw a line to the matching house



$$8 + 8$$



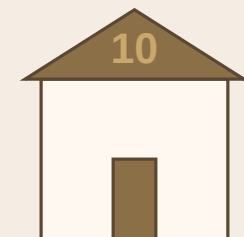
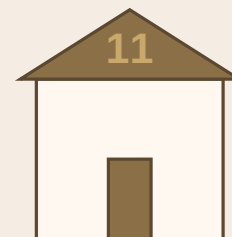
$$2 + 9$$



$$1 + 9$$



$$8 + 6$$



$$6 + 5$$



$$7 + 9$$



$$5 + 9$$



$$6 + 4$$



Abacus Calculation: Composition of 10 (+8)



+8 = +3 - 5 + 10 Fingering Ex.: 5+8, 6+8

S.No.	1	2	3	4	5	6	7	8
1	5	6	5	6	5	6	5	6
2	+8	+8	+8	+8	+8	+8	+8	+8
3	-2	-3	-1	-2	-3	-1	-2	-3
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	5	6	5	6	5	6	5	6
2	+8	+8	+8	+8	+8	+8	+8	+8
3	-1	-2	-3	-1	-2	-3	-1	-2
Ans.								

S.No.	1	2	3	4
1	5	6	5	6
2	+8	+8	+8	+8
3	-3	-1	-2	-3
Ans.				

5 + 8

= 5+3-5+10

= 13

13

5 + 8 =

6 + 8 =

5 + 8 =

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8
1	5	6	5	6	5	6	5	6
2	+8	+8	+7	+7	+6	+6	+5	+5
3	-3	-1	-2	-3	-1	-2	-3	-1
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	7	8	9	5	6	7	8	9
2	+8	+7	+6	+5	+4	+3	+2	+1
3	-2	-3	-1	-2	-3	-1	-2	-3
Ans.								

S.No.	1	2	3	4
1	5	6	7	8
2	+8	+7	+5	+4
3	-1	-2	-3	-1
Ans.				

+

 $7 + 6 =$

+

 $8 + 5 =$

+

 $5 + 8 =$

+

 $6 + 8 =$

Abacus Calculation: Composition of 10 (+9)



+9 = +4 - 5 + 10 Fingering Ex.: 5+9

S.No.	1	2	3	4	5	6	7	8
1	5	5	5	5	5	5	5	5
2	+9	+9	+9	+9	+9	+9	+9	+9
3	-2	-3	-1	-2	-3	-1	-2	-3
Ans.								

S.No.	1	2	3	4	5	6	7	8
1	5	5	5	5	5	5	5	5
2	+9	+9	+9	+9	+9	+9	+9	+9
3	-1	-2	-3	-1	-2	-3	-1	-2
Ans.								

S.No.	1	2	3	4
1	5	5	5	5
2	+9	+9	+9	+9
3	-3	-1	-2	-3
Ans.				

 +
 

5 + 9 =

 +
 

6 + 8 =

 +
 

7 + 7 =

5 + 9
 = 5+4-5+10
 = 14

14

Abacus Calculation: Revision



S.No.	1	2	3	4	5	6	7	8	9	10
1	1	6	7	3	6	5	7	6	8	3
2	+5	+7	+5	+6	+8	+7	+5	+3	+4	+6
3	+2	-4	+3	+5	-3	-6	-2	+5	-3	+2
Ans.										



S.No.	1	2	3	4	5	6	7	8	9	10
1	1	6	5	5	7	6	5	4	3	2
2	-1	-3	+8	+4	-2	+8	+7	+8	+9	+5
3	+4	+5	-2	-3	+6	-4	-3	-2	-1	+7
Ans.										

S.No.	1	2	3	4	5	6	7	8	9	10
1	4	7	6	5	7	3	2	6	5	7
2	+5	-2	+5	+4	+3	+7	+5	+4	+6	+8
3	-3	+8	-3	-2	+8	-2	+3	-3	-2	-3
Ans.										



How to Calculate When the Sum is More Than 9

Every digit 6, 7, 8, 9 is made of 5 plus a part



How to calculate
when the sum is
more than 9?

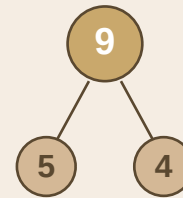
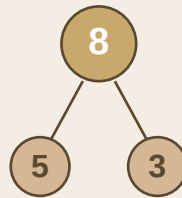
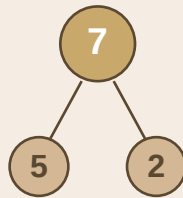
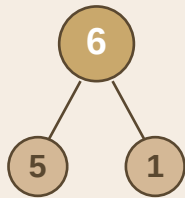
Examples:

$$5 + 6 = ?$$

$$5 + 7 = ?$$

$$5 + 8 = ?$$

$$5 + 9 = ?$$



★ Adding 6, 7, 8 and 9:

$$+ 6 = + 1 - 5 + 10$$

$$+ 7 = + 2 - 5 + 10$$

$$+ 8 = + 3 - 5 + 10$$

$$+ 9 = + 4 - 5 + 10$$



The Process of Mental Arithmetic

Picture the abacus in your mind — no fingers needed



Close your eyes

Imagine the abacus

Imagine the beam

Imagine the rods

Imagine the beads

$$1 + 2 + 5 = ?$$

$$\begin{array}{r} 1 \\ +2 \\ +5 \end{array}$$

$$4 - 1 + 3 = ?$$

$$\begin{array}{r} 4 \\ -1 \\ +3 \end{array}$$



Mental Calculation



S.No.	1	2	3	4	5	6	7	8	9	10
A	1	1	3	2	3	4	2	4	1	3
2	+1	+2	+1	-1	-1	-1	-2	-2	+2	-2
3	+2		-2	-1	-3	-1	-2	+1	-1	+1
Ans.										
Ans.										



Mental Calculation



S.No.	1	2	3	4	5	6	7	8	9	10
A	7	8	7	9	4	3	2	4	8	7
2	-6	+1	-5	-3	-2	+6	+4	+9	-1	-5
3	+5	-5	+2	+4	-1	-7	+1	-2	+5	+2
Ans.										
Ans.										

S.No.	1	2	3	4	5	6	7	8	9	10
B	6	2	4	8	6	4	2	8	2	8
2	+3	+2	+3	-6	-5	+2	+6	-4	-1	-7
3	-7	+5	+2	+8	+3	-3	-5	-2	+1	+2
Ans.										
Ans.										

Listening Exercise

1 _____

2 _____

3 _____

4 _____

5 _____

6 _____

7 _____

8 _____

9 _____

10 _____

Mental Calculation



S.No.	1	2	3	4	5	6	7	8	9	10
A	7	8	2	4	8	5	1	6	9	7
2	-6	+1	+7	-3	+5	+3	+4	+5	-2	-5
3	+5	-5	-7	+3	-7	-5	+5	-7	-6	+2
Ans.										
Ans.										

S.No.	1	2	3	4	5	6	7	8	9	10
B	6	8	7	3	6	5	1	7	4	6
2	+3	-3	-4	+4	-5	+2	+6	-5	+3	-2
3	-7	+2	+3	-6	+7	-5	-5	-2	-4	+2
Ans.										
Ans.										

Listening Exercise

1 _____

2 _____

3 _____

4 _____

5 _____

6 _____

7 _____

8 _____

9 _____

10 _____

Mental Calculation



S.No.	1	2	3	4	5	6	7	8	9	10
A	2	5	3	5	8	3	5	4	7	4
2	+2	+3	-2	+3	+5	-8	+1	+4	-6	+5
3	+5	-2	+6	-8	-7	+5	-2	-3	+8	-7
Ans.										
Ans.										



S.No.	1	2	3	4	5	6	7	8	9	10
B	5	2	6	5	7	4	6	9	8	3
2	+2	+5	+5	+4	-3	+2	+6	-5	-7	+4
3	+1	-5	-5	-2	+5	-3	-5	-2	+1	+6
Ans.										
Ans.										



Listening Exercise

1 _____

2 _____

3 _____

4 _____

5 _____

6 _____

7 _____

8 _____

9 _____

10 _____

Mental Calculation



S.No.	1	2	3	4	5	6	7	8	9	10
A	1	6	4	9	2	1	3	5	8	7
2	+6	-5	-2	-7	+7	+5	+1	+4	-2	-6
3	+4	-2	+3	+2	-7	+2	+6	-6	-6	+2
Ans.										
Ans.										

S.No.	1	2	3	4	5	6	7	8	9	10
B	7	3	9	8	3	2	3	8	4	6
2	+1	+5	-4	-6	+5	+2	+6	-6	-2	-2
3	-6	-4	+3	+2	-7	+5	-5	+1	+1	+3
Ans.										
Ans.										

Listening Exercise

1 _____

2 _____

3 _____

4 _____

5 _____

6 _____

7 _____

8 _____

9 _____

10 _____